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FreqTransNet: A Frequency-aware Transformer Network for Robust Image Watermarking

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***Abstract*—To conquer the limitations of traditional watermarking algorithms in adaptively embedding watermarks and to address the constrained feature representation capability of convolution-based methods, this paper introduces a robust deep learning watermarking model named FreqTransNet. This model effectively integrates convolutional modules, Transformer structures, and frequency-domain transformations to enhance its ability in both capturing multi-scale image features and improving robustness. Specifically, the self-attention mechanism inherent in the Transformer effectively models long-range dependencies and global context, thereby improving the model's representation of complex structures. The integration of frequency-domain transformations also mitigates spatial distortions, further amplifying resistance to interference. The extensive experimental results reveal that FreqTransNet surpasses existing state-of-the-art deep learning watermarking approaches in both visual quality and robustness. Compared to the best-performing baseline, FreqTransNet achieves a notable 3.09 dB enhancement in PSNR (from 47.46 dB to 50.55 dB). Additionally, it consistently maintains extraction accuracy exceeding 97% in the presence of various attack scenarios, highlighting its practical applicability and stability in real-world applications.**

***Index Terms*—Robust watermarking; Transformer; Fast Fourier transform.**

I. INTRODUCTION

DIGITAL watermarking technology, recognized as a proactive authentication method, has been extensively utilized in the fields of copyright protection and integrity verification [1]-[3]. By invisibly embedding watermark information within the host image, this technique facilitates effective tracking of image ownership. Even in the face of significant distortions during image transmission, it is often feasible for the receiver to extract the embedded watermark with minimal or no degradation. The performance of watermarking algorithms is typically assessed using two primary metrics: robustness and imperceptibility.

Watermarking algorithms can generally be classified into two categories based on embedding capacity: zero-watermarking and non-zero-watermarking schemes. The zero-watermarking algorithms do not alter the original image data during the embedding process; instead, they extract essential features from the image and fuse these with the watermark to create a unique feature fingerprint. This approach maintains the quality of the original image while ensuring copyright protection [4]-[6]. Conversely, non-zero-watermarking algorithms embed watermark information by selectively altering salient or stable regions of the image, thereby enhancing robustness [7]-[8]. Despite the differing embedding strategies, both types of algorithms fundamentally depend on the representation of image features.

Generally, a diverse array of methods exists for selecting image features. A commonly employed strategy involves applying a spatial-frequency domain transformation to the host image, followed by the embedding of watermark information into the resultant frequency-domain coefficients. These methods leverage the stability of frequency components, thus improving both the robustness and imperceptibility of the watermarking algorithm. This group of techniques encompasses several prominent methodologies, including the discrete Fourier transform (DFT) [9], discrete wavelet transform (DWT) [10], and discrete cosine transform (DCT) [11]. Such transformations enable the embedding of watermark information into critical frequency regions of the image, minimizing perceptual distortion while enhancing resistance against compression, filtering, and other attacks. However, these strategies often depend on manually designed embedding techniques and strength parameters, which may limit their adaptability to varying image distributions or attack scenarios.

In addition to frequency-domain techniques, watermark information can also be embedded into geometric invariants of the host image, such as Trinion fractional-order continuous orthogonal moments [12], polar harmonic Fourier moments [13], and Bessel-Fourier moments [14]. These orthogonal moments demonstrate strong invariance to rotation and translation, thereby offering enhanced robustness against geometric attacks. To enrich feature representation and capture finer image details, higher-order moments are often utilized. Nevertheless, higher-order moments exhibit susceptibility to the "central distortion" phenomenon during computation, where moment values near the image center are particularly sensitive to minor perturbations, adversely impacting overall robustness [13]-[15]. This limitation somewhat restricts their applicability in complex watermarking tasks.

In recent years, deep learning methods have been

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increasingly adopted for digital watermarking tasks due to their superior feature extraction capabilities and efficient end-to-end optimization frameworks. Initial approaches predominantly employed convolutional neural networks (CNNs) [16]-[17] and deep neural networks (DNNs) [18]-[20]. Notably, HiDDeN [21] was the first to propose an end-to-end watermarking framework, paving the way for subsequent advancements in network architectures and encoding strategies. For instance, the two-stage deep learning (TSDL)-based approach [22] incorporated perceptual loss and dynamic noise simulation to enhance model robustness against common attacks. Compared to traditional digital watermarking algorithms, deep learning-based methodologies present noticeable advantages in adaptive embedding and imperceptibility. Presently, the encoder-noise-decoder (END) framework has emerged as the leading architecture, enabling unified watermark embedding and extraction.

In addition to end-to-end frameworks, some researchers have investigated flow-based watermarking models [23]. These models utilize invertible architectures to achieve high-fidelity embedding and robust extraction, effectively balancing imperceptibility and robustness. However, such methods often suffer from structural complexity and unstable training processes. A brief comparison between the END framework and flow-based watermarking models is illustrated in Fig. 1.

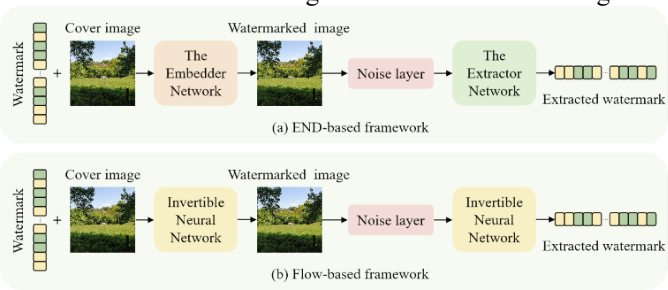


Fig.1. The network comparison of END and flow-based framework.

With the continuous evolution of DNN-based architectures, research has gradually shifted from simple convolutional stacking to more sophisticated and expressive structural designs. The transformer architectures, utilizing self-attention mechanisms, enable global feature modeling across the entirety of an image. This innovation addresses the limitations inherent in traditional convolutional networks, such as restricted receptive fields and dependence on local context, thereby unlocking new avenues to enhance the robustness and generalization capabilities of watermarking algorithms in complex scenarios [24]–[26]. Although Transformers excel at capturing global dependencies, they are commonly challenged by high computational complexity, sensitivity to positional information, and a tendency to neglect fine-grained local details, which can limit their effectiveness in certain vision tasks.

To address these limitations, this paper introduces a robust image watermarking framework, the so-called FreqTransNet. Built upon the classic END architecture, FreqTransNet supports adaptive embedding and extraction of watermark information, thereby augmenting the overall flexibility and scalability of the system. Within the encoder design, FreqTransNet synergistically integrates fast Fourier transform (FFT) modules with self-attention mechanisms to simultaneously capture frequency-domain characteristics and spatial contextual information. This fusion strategy not only enhances robustness against a broad spectrum of attacks but also ensures the preservation of high visual quality. Furthermore, the performed experimental results demonstrate that FreqTransNet consistently outperforms existing state-of-the-art (SOTA) methods across various attack scenarios.

Briefly, our main contributions can be summarized as follows:

1) We introduce FreqTransNet, an innovative watermarking network architecture that effectively balances fidelity to the original image and robustness against various distortions.
2) FreqTransNet integrates a global self-attention mechanism alongside a frequency enhancement module, significantly enhancing its adaptability to complex images while bolstering robustness against common attacks, including noise, compression, and geometric distortions.
3) The experimental findings demonstrate that FreqTransNet surpasses existing deep learning-based watermarking methodologies in terms of both visual quality and robustness.

The structure of this paper is organized as follows: Section II provides a review of the background and related work. Section III details the theoretical framework and the implementation of the proposed network. Section IV presents the experimental evaluations, while Section V concludes the discussion.

II. Related Work

A. Traditional robust watermarking

A typical digital watermarking system comprises three primary stages: watermark generation, embedding, and extraction. To achieve robustness against common signal distortions, traditional watermarking methods primarily focus on how and where watermark information is embedded into the host image. From a methodological standpoint, robust watermarking strategies can be classified into two main categories: frequency-domain embedding methods, which integrate watermark information into transform-domain coefficients, and invariant-feature-based methods, which exploit stable image descriptors to withstand geometric distortions.

In the realm of frequency-domain embedding, various transform techniques have been utilized to bolster robustness. Nawaz et al. [27] introduced a zero-watermarking scheme tailored for medical images utilizing the DCT to facilitate watermark authentication through the manipulation of frequency-domain coefficients, supplemented by auxiliary feature extraction mechanisms to improve robustness against diverse attacks. In a contrasting approach, Begum et al. [28] capitalized on the multi-resolution capabilities of the DWT by proposing a blind watermarking technique that integrates

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watermark information into selected wavelet sub-bands, thus achieving a balance between robustness and imperceptibility. Extending the application of wavelet-domain methods, Luo et al. [29] formulated a watermarking framework grounded in the integer wavelet transform (IWT), where multi-scale integer wavelet coefficients act as reliable embedding carriers, ensuring efficient watermark extraction while maintaining high visual quality.

Beyond frequency-domain embedding, another significant research trajectory focuses on watermarking techniques predicated on geometric invariant features. These methods create watermarks using descriptors invariant to rotation and translation, thereby enhancing resistance to geometric distortions. Within this category, He et al. [30] shifted focus from merely embedding strategies to improving moment accuracy, proposing a watermarking scheme leveraging polar harmonic Fourier moments (PHFMs), whereby precise moment computation secures stable watermark construction. In pursuit of computational efficiency, Wang et al. [31] introduced a zero-watermarking methodology based on the fast fractional-order polar linear canonical transform, coupled with an asymmetric tent map. While preceding moment-based techniques predominantly targeted grayscale images, practical watermarking implementations frequently necessitate addressing color images. To meet this demand, quaternion theory has been integrated into invariant moment–based watermarking frameworks. Sheng et al. [32] advanced moment-based watermarking for color medical images by proposing a zero-watermarking scheme founded on fast quaternion radial harmonic Fourier moments, demonstrating considerable robustness against desynchronization attacks.

B. Deep learning-based robust watermarking

A recent comprehensive survey on deep learning–based image watermarking systematically reviews existing methodologies from the perspectives of network architecture, system organization, and evaluation criteria [33]. In recent years, researchers have progressively introduced watermarking algorithms leveraging various deep learning models [16-17, 18-20, 24-26]. Notably, in 2018, Zhu et al. [21] were pioneers in employing three convolutional networks for data hiding, presenting the classical HiDDeN model, which exhibited substantial robustness against a diverse array of image processing attacks. In this regard, Li et al. [16] integrated traditional frequency-domain transformations with neural networks to propose a CNN-based collaborative architecture that embeds grayscale watermarks into the DCT domain utilizing deep learning techniques. Similarly, Lee et al. [17] introduced a CNN-based blind watermarking framework that adapts to both image resolution and watermark information, attaining superior performance across various attacks.

Zhong et al. [34] developed a highly adaptive deep neural network framework that facilitates robust watermark embedding and extraction in an unsupervised manner without reliance on prior knowledge. Li et al. [35] incorporated skip connections within the network, enabling the sharing of feature information from decoder layers with the encoder, thus enhancing inter-module correlation. Zhao et al. [36] proposed DARI-Mark, utilizing joint modeling of channel and spatial attention mechanisms to identify non-salient regions within the image and assigning them higher weights. This approach allows for adaptive optimization during watermark embedding, significantly improving imperceptibility. Liao et al. [37] introduced adaptive payload distribution strategies for multi-image steganography, allocating embedding payloads based on image texture characteristics, which enhances security against contemporary steganalysis.

Continuing the trend towards more robust global feature modeling, Transformer-based models are increasingly incorporated into watermarking frameworks to enhance global feature representation. Un et al. [24] introduced the Swin-Unet structure into the watermark embedding process, bolstering both global modeling capabilities and robustness against proxy-based attacks. Luo et al. [25] developed a robust watermarking framework named WFormer, integrating cross-domain attention fusion and feature enhancement modules to withstand complex perturbations. Wu et al. [26] designed a collaborative network, CEWformer, which jointly optimizes imperceptibility and robustness by combining image enhancement with watermark embedding.

As deep watermarking models continue to advance in both structure and capacity, their range of application scenarios has significantly broadened. In this regard, Fu et al. [38] introduced a robust watermarking framework that incorporates wavelet-domain cascaded embedding and recovery mechanisms, aiming to enhance watermark fidelity, particularly in the presence of screen-shooting attacks. Li et al. [39] further tackled the challenges posed by practical screen-camera distortions by proposing a screen-shooting-resistant watermarking framework. This approach is built on a simulation of screen-shooting distortions, which explicitly models the screen-camera process to bolster robustness and visual quality in real-world leakage scenarios. Extending beyond image watermarking, Tancik et al. [40] developed StegaStamp, which simulates realistic distortions from printing and photography, thus facilitating reliable information extraction from video frames affected by dynamic lighting changes, occlusion, and other complex variations. In a related forensic context, Liao et al. [41] introduced a facial muscle motion-based (FAMM) framework aimed at detecting compressed Deepfake videos by modeling the temporal dynamics of facial landmarks, achieving notable robustness under the compression typically encountered on social networks.

III. Proposed Architecture

We propose an END robust watermarking framework based on Lite spatial transformer and FFT, called FreqTransNet, and the corresponding flowchart is presented in Fig.2.

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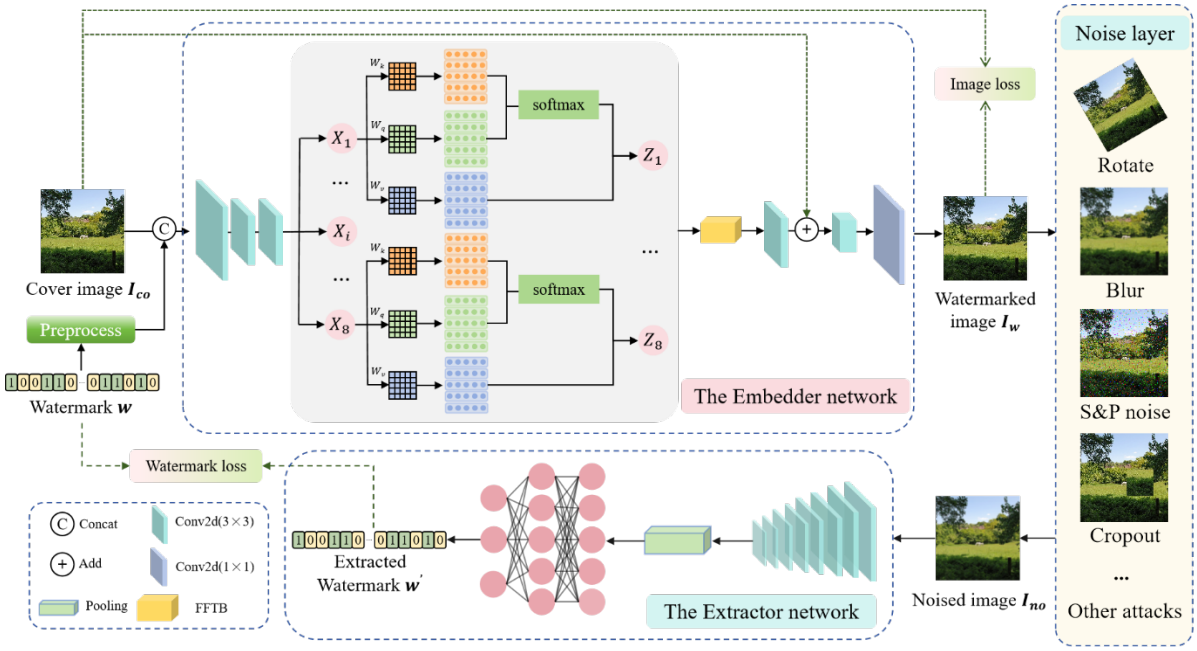


Fig.2. The framework of FreqTransNet. It consists of three main parts: the encoder, the noise layer and the decoder.

A. The embedder network

The embedding network proposed herein consists of two main components: the encoding module and the embedding module. The encoding module first expands the watermark vector $w \in \mathbb{R}^L$ to match the spatial resolution of the cover image $I_{co} \in \mathbb{R}^{H \times W \times 3}$, enabling effective concatenation and fusion along the channel dimension. Afterward, the original image undergoes three convolutional operations, followed by a feature extraction module that further models the image's semantic representations. Finally, the watermark, original image, and encoded features are concatenated and fused, and three additional convolutional layers are applied to produce the watermarked image. Taking $I_{co} \in \mathbb{R}^{256 \times 256 \times 3}$ and $w \in \mathbb{R}^{64}$ as an example, the detailed pipeline is illustrated in Fig.3.

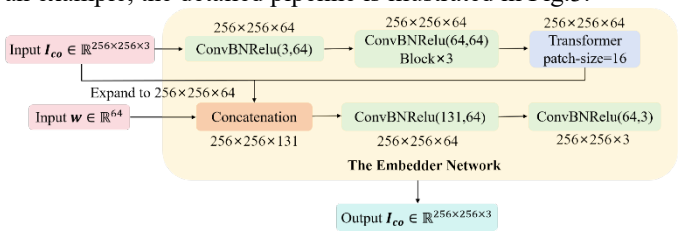


Fig.3. Detailed pipeline of the proposed embedder network (example: 256×256 image and 64-bit watermark)

To minimize the reconstruction error between I_w and I_{co}, E is trained to iteratively adjust θ_{em} to preserve visual similarity and embedding transparency, following the loss $\mathcal{L}_{em}$.

$$\mathcal{L}_{em} = MSE(I_{co}, I_w) = MSE(I_{co}, E(\theta_{em}, I_{co}, w)) \quad (1)$$

where $MSE(\cdot)$ represents the mean square error function.

B. Noise layer

To enhance the robustness of the watermarking system against various distortions, a perturbation module is integrated during the training phase, serving as a critical component for improving the robustness of the framework.

The present investigation encompasses a comprehensive range of distortion types specifically designed to simulate real-world attack scenarios. The geometric distortions include Cropout, Dropout, and Scaling, while non-geometric distortions encompass Gaussian noise, Salt & Pepper noise, Gaussian blur, Median blur, and JPEG compression.

During the training process, the perturbation module applies varying degrees of distortion to the embedded image, resulting in a noised version that is subsequently processed by the extractor network. This approach empowers the network to uphold effective watermark extraction performance amidst complex and diverse attack conditions. Consequently, the types and magnitudes of distortions configured in this module directly impact the system's overall robustness against various attack vectors. Fig. 4 illustrates the noise layer's operation when a combined attack is applied.



Fig.4. The pipeline of the noise layer (example: combined noise)

C. The extractor network

The extraction network ex is architecturally similar to the embedding network and can be divided into extraction and decoding modules. In practical applications, the input to the extraction network may comprise either the original watermarked image $I_w \in \mathbb{R}^{H \times W \times 3}$ or the attacked watermarked image $I_{no} \in \mathbb{R}^{H \times W \times 3}$. The input image undergoes multiple convolutional layers, followed by an adaptive average pooling layer and a fully connected layer, ultimately producing the estimated watermark vector w'. Taking $I_{no} = \mathbb{R}^{256 \times 256 \times 3}$ as an illustrative example input, the detailed processing pipeline is depicted in Fig.5.

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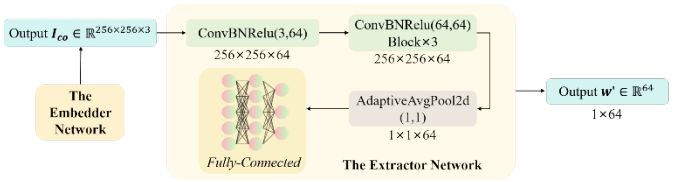


Fig.5. Detailed pipeline of the proposed extractor network (example: 256×256 image and 64-bit watermark)

The objective of ex is to iteratively optimize its parameters θ_{ex} such that the extracted watermark w' closely approximates the original watermark message w, effectively minimizing the reconstruction error between them, following the loss $\mathcal{L}_{ex}$.

$$\mathcal{L}_{ex} = MSE(M, M') = MSE(M, D(\theta_{ex}, M, M')) \tag{2}$$

D. Loss Function

In image watermarking systems, the design of the loss function is crucial for achieving a suitable balance between imperceptibility and robustness. FreqTransNet employs a joint optimization objective that incorporates both image reconstruction loss and decoding loss, which can be formulated by:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{em} + \lambda_2 \mathcal{L}_{ex} \tag{3}$$

where λ_1 and λ_2 are manually tunable hyperparameters. It is worth mentioning that a larger value of λ_1 encourages the network to prioritize visual fidelity of the watermarked image, whereas a larger λ_2 places greater emphasis on enhancing the robustness of watermark extraction. In practice, these hyperparameters can be adjusted flexibly according to meet the specific requirements of various application scenarios.

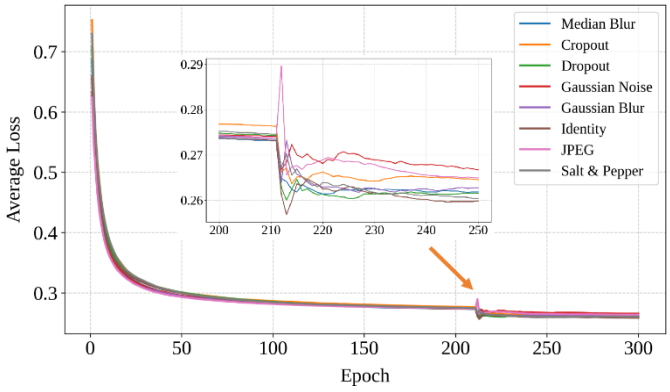


Fig.6. Loss Variation of **FreqTransNet**

Fig.6 illustrates the loss variation of FreqTransNet across both non-attack conditions and seven distinct attack scenarios. Although fluctuations are evident between epochs 200 and 250—typically attributed to instability in the optimizer's update process—the subsequent consistent decline in loss signifies that the model is progressively converging, thereby demonstrating the efficacy of the training process.

IV. EXPERIMENTS AND ANALYSIS

A. Implementation Details

Dataset: The whole network was trained on the COCO dataset [42], from which we selected 10,000 images for the training phase. The resolution was uniformly scaled, and 10,000 binary CIFAR images served as watermarks. The proposed methodology effectively learns the watermarking algorithm while avoiding overfitting.

Metrics: During the testing phase, we employ the PSNR, SSIM, and extraction bit accuracy (ACC) to quantitatively evaluate visual quality and robustness. In particular, the ACC factor is defined as

$$ACC = \frac{1}{L}\sum_{i=1}^{L} \mathbb{1}[w_i = w_i^*], \tag{4}$$

where L denotes the watermark length, w_i represents the embedded bit, w_i^* is the extracted bit, and $\mathbb{1}[\cdot]$ stands for the indicator function.

Benchmarks: To demonstrate the superiority of the proposed method, we conduct a comparative analysis with several SOTA image watermarking techniques, including HiDDeN [21], TSDL [22], DADW [43], and MBRS [44]. For the robustness evaluation, we examine five non-geometric distortions: Gaussian Noise (GN), Salt-and-Pepper Noise (SPN), Gaussian Blur (GB), Median Blur (MB), and JPEG Compression (JPEG), alongside three geometric attacks: Cropout (CP), Dropout (DP), and Scaling (SC). To ensure a thorough assessment of robustness, a dedicated watermarking network is trained for each type of distortion. For the sake of fairness in comparison, all methods are trained on the same dataset using identical noise settings. All experiments are conducted on images sized at 256×256, with the watermark message length standardized to 64 bits.

Platforms: The framework is implemented in PyTorch [45], with all experiments carried out on a PC equipped with a 2.60 GHz Intel Core i7 CPU, NVIDIA RTX 4060Ti, and 24GB of RAM.

B. Visual Quality and Robustness

This section presents simulation results under eight types of image distortions, comparing the visual quality and robustness of the proposed method with other existing approaches. As illustrated in Fig.7, the first to third rows display the cover image I_{co}, the embedded image I_{em}, and the distorted image I_{no}, respectively, whereas the fourth row represents the residuals between I_{em} and I_{no}. These residuals vary with the distortion types, indicating the model's adaptability and strong representation capability.

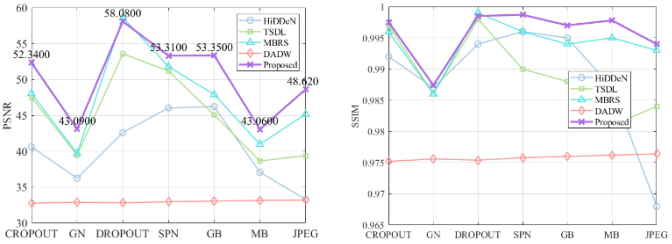


Fig.7. Visualization and residual analysis of watermarked images under multiple distortion types

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For imperceptibility evaluation, Fig. 8 displays the PSNR and SSIM comparisons with several SOTA methods under various distortions. The results demonstrate the superior visual quality preservation of our methodology during watermark embedding.



(a) The PSNR of the embedded images with different methods

(b) The SSIM of the embedded images with different methods

Fig.8. Comparison of image quality metrics under various distortions

1) *Cropout distortion:* Cropout involves replacing portions of the watermarked image with corresponding regions from the original host image (i.e., before watermark embedding). During the training phase, Cropout distortion was implemented with a fixed intensity of 40%. Table I summarizes the experimental results across various cropping ratios ranging from 10% to 40%.

TABLE I

EXTRACTION ACCURACY WITH DIFFERENT CROPOUT RATIOS.

Model	ACC (%)↑			
	r=0.1	0.2	0.3	0.4
HiDDeN [21]	95.63	94.73	88.75	76.88
TSDL [22]	98.72	98.54	96.88	93.75
DADW [43]	92.01	91.67	91.44	90.26
MBRS [44]	99.71	**99.22**	97.18	90.43
Proposed	**99.83**	99.16	**98.79**	**98.53**

The proposed approach consistently exhibits notable robustness while preserving high visual fidelity. Significantly, for cropping ratios between 10% and 40%, the watermark extraction accuracy remains above 97%, attesting to the methodology's robustness against cropout attacks.

2) *Dropout distortion:* Dropout distortion shares operational characteristics with Cropout distortion, as both simulate content loss by modifying pixel values within the image. However, the primary distinction lies in their spatial patterns: Cropout replaces a specific contiguous region of pixels, whereas Dropout distortion randomly selects individual pixels throughout the image for replacement, resulting in more unpredictable and dispersed distortions.

TABLE II

EXTRACTION ACCURACY WITH DIFFERENT DROPOUT RATIOS.

Model	ACC (%)↑				
	r=0.2	0.3	0.4	0.5	0.6
HiDDeN [21]	90.21	89.58	87.08	86.74	82.71
TSDL [22]	97.54	95.21	93.54	92.29	90.42
DADW [43]	90.60	90.04	87.71	83.96	80.54
MBRS [44]	96.29	94.73	94.15	92.58	90.63

During training, the dropout ratio is fixed at 0.4. For testing, the ratio is varied between 0.2 and 0.6 to thoroughly assess the model's robustness under varying distortion levels, as detailed in Table II. When compared to prevailing SOTA watermarking methods, the proposed FreqTransNet demonstrates enhanced robustness against Dropout distortion. Remarkably, it maintains high extraction accuracy even under severe distortion conditions, highlighting its strong robustness to random content loss.

3) *Gaussian noise:* Gaussian noise represents one of the most common distortions encountered during image transmission. To enhance noise robustness, the training process incorporated Gaussian perturbations with variances randomly sampled from the range [0.001, 0.04]. During testing, a broader variance interval of [0.01, 0.05] was utilized to evaluate the model's robustness under varying noise intensities, with comprehensive results outlined in Table III.

TABLE III

EXTRACTION ACCURACY WITH DIFFERENT RATIOS OF GAUSSIAN NOISE.

Model	ACC (%)↑				
	r=0.01	0.02	0.03	0.04	0.05
HiDDeN [21]	89.58	86.46	83.96	83.12	79.17
TSDL [22]	92.08	91.25	88.33	87.08	82.92
DADW [43]	68.75	60.63	60.00	57.71	56.04
MBRS [44]	**99.91**	99.42	98.10	96.09	94.15
Proposed	99.89	**99.62**	**98.96**	**98.56**	**97.32**

As observed in Table III, the proposed method consistently outperforms existing approaches in watermark extraction, demonstrating strong robustness against Gaussian noise distortion.

4) *Salt & Pepper noise:* Similar to Gaussian noise, salt-and-pepper noise is another common type of random distortion encountered during image transmission. It is characterized by the random replacement of a percentage of pixels with extreme values (black or white). During the training phase, the noise ratio was randomly selected within the range of 0.001 to 0.04, while in the testing phase, it varied from 0.01 to 0.05 to evaluate the model's robustness across various distortion levels, as detailed in Table IV.

TABLE IV

EXTRACTION ACCURACY WITH DIFFERENT RATIOS OF SALT&PEPPER NOISE.

Model	ACC (%)↑				
	r=0.01	0.02	0.03	0.04	0.05
HiDDeN [21]	95.12	93.79	93.45	92.92	90.42
TSDL [22]	97.29	95.63	93.54	92.71	91.46
DADW [43]	90.42	86.67	80.42	75.42	71.42
MBRS [44]	98.05	98.74	98.34	97.56	96.68
Proposed	**99.77**	**99.12**	**98.56**	**97.88**	**97.34**

The experimental results demonstrate that the proposed approach consistently achieves extraction accuracy exceeding 98% across all noise intensities, indicating strong robustness against salt-and-pepper noise interference.

5) *Gaussian blur:* For Gaussian blur distortion, the noise layer was configured with a fixed variance of 2 during training to enhance robustness against significant blurring effects. In the testing phase, a Gaussian blur was applied with variances ranging from 0 to 2 to assess performance under varying degrees of distortion. The extraction accuracy is illustrated in Table V.

TABLE V

EXTRACTION ACCURACY WITH DIFFERENT VARIANCE OF GAUSSIAN BLUR.

Model	ACC (%)↑			
	r=0.00001	0.5	1	2
HiDDeN [21]	95.44	95.21	94.33	84.37
TSDL [22]	**99.92**	**99.79**	98.48	93.21
DADW [43]	91.70	89.80	76.25	66.25

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MBRS [44]	98.64	98.25	97.66	87.80
Proposed	98.99	99.16	**98.96**	**97.34**

6) *Median blur:* Median filtering is a broadly employed technique for image smoothing. To enhance the model's robustness, a fixed 7×7 filter was utilized during training, while robustness evaluation was conducted with a 3×3, 5×5, and 7×7 window sizes. The experimental results are summarized in Table VI.

TABLE VI
EXTRACTION ACCURACY WITH DIFFERENT VARIANCE OF MEDIAN BLUR.

Model	ACC (%)↑		
	3 × 3	5 × 5	7 × 7
HiDDeN [21]	86.25	83.70	79.71
TSDL [22]	99.38	97.21	95.12
DADW [43]	73.15	72.71	60.63
MBRS [44]	**99.42**	**98.93**	97.27
Proposed	99.31	98.65	**97.53**

As illustrated in Table VI, the proposed method consistently maintains superior robustness across all tested blur levels. Notably, it achieves extraction accuracies surpassing 98% under all median filter sizes, highlighting its strong robustness against median blur distortions.

7) *JPEG compression*: JPEG compression is one of the most common operations in image storage and format conversion, and representing a typical non-geometric distortion that affects watermark robustness. During the training phase, the quality factor (QF) is set to 50 as the default configuration, while in the testing phase, the model is evaluated against varying compression levels with QF ranging from 40 to 90. The corresponding results are presented in Table VII.

TABLE VII
EXTRACTION ACCURACY WITH DIFFERENT VARIANCE OF JPEG COMPRESSION.

Model	ACC (%)↑					
	40	50	60	70	80	90
HiDDeN [21]	86.67	91.24	92.92	93.33	93.54	94.38
TSDL [22]	91.04	91.46	93.96	94.21	94.35	94.74
DADW [43]	72.25	73.32	74.37	77.92	81.05	86.46
MBRS [44]	94.83	94.93	96.68	97.66	97.66	98.84
Proposed	**95.66**	**96.22**	**97.56**	**98.77**	**99.56**	**99.95**

As indicated in Table VII, the proposed method consistently achieves robust watermark extraction performance across various JPEG compression levels. Notably, at the typical compression level of QF=50, the extraction accuracy exceeds 96%, demonstrating the method's practical applicability in real-world compression scenarios. Specifically, by explicitly incorporating spectral characteristics during embedding, the proposed design effectively preserves watermark-related features under JPEG quantization.

8) *Scaling:* To enhance the model's adaptability to varying image sizes, random scaling was applied during the training phase, with scaling factors ranging from 0.7 to 1.8 times the original image size. During the testing phase, four distinct scaling factors were employed to comprehensively evaluate the model's robustness to size variations.

TABLE VIII
EXTRACTION ACCURACY WITH DIFFERENT SCALING RATIOS.

Model	ACC (%)↑			
	r=0.5	1	1.5	2
HiDDeN [21]	90.31	95.93	92.98	91.02
TSDL [22]	75.03	82.30	85.96	75.90
MBRS [44]	82.08	92.79	88.94	82.88
Proposed	**97.36**	**99.42**	**98.12**	**99.07**

FreqTransNet consistently achieved extraction accuracy above 97% in robustness tests, demonstrating its exceptional practicality and reliability in handling scaling distortions. This result highlights the model's robustness in the face of varying scales and substantiates its effectiveness in real-world applications.

In addition to the representative baseline methods commonly adopted in earlier studies, this work further broadens the comparison to incorporate more recent learning-based watermarking approaches to assess the proposed method against contemporary robustness benchmarks.

TABLE IX
EXTRACTION ACCURACY COMPARISON WITH RECENT STUDIES.

Attacks	Proposed	Ref. [46]	Ref. [47]	Ref. [48]
Gaussian Noise 0.01	**99.89**	-	99.82	96.93
Gaussian Noise 0.02	**99.62**	99.20	-	-
Gaussian Noise 0.05	**97.32**	77.93	-	-
Salt&Pepper Noise 0.05	97.34	**99.22**	-	-
Median Blur 3 × 3	99.31	**99.81**	-	-
Cropout 0.1	**99.83**	99.22	-	-
Scaling 0.5	97.36	**99.22**	-	-
JPEG 90	99.95	**100.00**	99.80	90.40
JPEG 80	99.56	-	**99.81**	90.63
JPEG 70	98.77	-	**99.82**	90.57
JPEG 60	97.56	-	**99.74**	90.23
JPEG 50	96.22	**99.90**	99.66	90.27
Dropout 0.2	**99.86**	-	65.58	99.13
Dropout 0.3	**99.57**	-	71.07	99.17
Dropout 0.4	**99.25**	-	76.71	99.20
Dropout 0.5	98.54	-	82.26	**99.00**
Dropout 0.6	98.14	-	86.74	**99.33**

Table IX presents a comparison of the extraction accuracy of the proposed method alongside three recent representative approaches across various attack scenarios. Overall, the proposed method demonstrates consistently high robustness across different types of distortions. Notably, it exhibits clear advantages under Gaussian noise and dropout attacks, where the extraction accuracy remains stable even at elevated levels of attack intensity.

Regarding JPEG compression, all methods display competitive performance; however, the proposed approach maintains almost stable accuracy as the compression strength increases. These findings are clearly indicative of the fact that the proposed method achieves a favorable robustness-stability trade-off across diverse attack conditions.

C. Social-Media Recompression Evaluation

In practical image distribution scenarios over online social networks (OSNs), watermarked images are often re-encoded by social-media platforms. Such platform-level upload-download pipelines typically entail aggressive recompression (and potential resizing), which introduces complex artifacts that go beyond standard synthetic distortions. To better reflect real-world conditions, additional experiments were conducted using a self-constructed OSNs recompression dataset. Specifically, for each OSN transmission setting, 500 watermarked images were collected through actual upload-download pipelines, from which 400 images were utilized for fine-tuning based on the JPEG-trained model, while the remaining 100 images were reserved for evaluation. The

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results reported in Table X represent average values derived from the 100 test images.

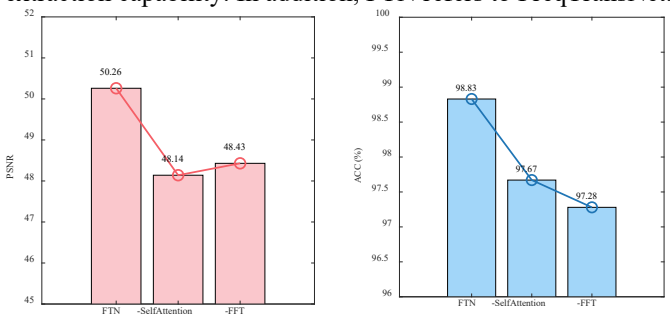
TABLE X
ROBUSTNESS AGAINST OSN RECOMPRESSION.

OSNs		PSNR	SSIM	ACC↑
1st	2nd			
WeChat	-	38.64	0.9416	95.25
WeChat	WeChat	36.92	0.9158	93.74
WeChat	Facebook	35.88	0.9021	91.63
Facebook	-	39.21	0.9484	95.87
Facebook	Facebook	37.05	0.9217	93.12
Facebook	WeChat	35.73	0.8975	90.21

Table X outlines the extraction accuracy under social-media recompression, revealing that the proposed framework retains robustness in the face of real platform-induced compression artifacts.

D. Ablation Studies

To validate the effectiveness of the individual components within the proposed FreqTransNet, ablation experiments were conducted, focusing on assessing the contributions of the self-attention mechanism and frequency domain transformation. The specific experimental results, as illustrated in Fig. 9, are based on the average values obtained from multiple attack scenarios, where the attack intensity remained consistent across all experiments. The abbreviations -SelfAttention and -FFT denote the removal of the self-attention mechanism and frequency domain transformation module, respectively, and serve to evaluate their impacts on both robustness and extraction capability. In addition, FTN refers to FreqTransNet.



(a) Ablation study (PSNR) (b) Ablation study (ACC)

Fig.9. Ablation results of FreqTransNet

1) *Effectiveness of the Self-Attention Mechanism:* The removal of the self-attention mechanism resulted in a reduction of 2.12 dB in PSNR and a 1.17% decrease in ACC for FreqTransNet. This finding highlights the critical role that the self-attention mechanism plays in capturing the global features of the image, thereby significantly enhancing both image quality and extraction accuracy.

2) *Effectiveness of the Frequency Domain Transformation:* When the frequency domain transformation module was omitted, FreqTransNet exhibited a decline of 1.83 dB in PSNR and a 1.56% decrease in ACC. This suggests that the frequency information facilitated by the frequency domain transformation (FFT) is vital for improving the model's performance.

E. Resolution Sensitivity Analysis

In practical watermarking implementations, it is common for platform-side systems to standardize user images to a uniform resolution prior to the embedding of watermarks. This practice enhances processing efficiency and supports lightweight deployment. Consequently, the proposed method is primarily evaluated using images at a resolution of 256×256 pixels, which serves as a typical operational standard in such deployment scenarios.

While the default configuration of the proposed framework utilizes an image resolution of 256×256, the framework is designed for seamless scalability to accommodate higher-resolution inputs. In light of real-world deployment considerations, experiments conducted in this section utilize 512×512 images as a representative higher-resolution benchmark. To ensure a fair comparison, the embedding rate is maintained consistently across different resolutions.

TABLE XI
RESOLUTION SENSITIVITY ANALYSIS OF PROPOSED METHOD.

	Resolution 256×256	Resolution 512×512
Imperceptibility		
PSNR (dB)	50.26	49.87
Robustness (ACC)		
Cropout 0.2	99.16	98.57
Dropout 0.4	99.23	98.63
Gaussian Noise 0.03	98.96	98.72
Salt&Pepper Noise 0.03	98.56	97.74
Gaussian Blur 0.5	99.16	98.92
Median Blur 5 × 5	98.65	97.96
JPEG 60	97.56	97.23
Scaling 1.5	98.12	97.68

Table X presents the extraction accuracy of the proposed framework at the two distinct image resolutions when subjected to various representative attacks. For each category of attack, a moderate intensity level is chosen for evaluation. The results demonstrate that the proposed framework exhibits consistent robustness when applied to higher-resolution images, thereby confirming its effective scalability.

V. CONCLUSION

In this paper, we establish a newly developed robust image watermarking model, the so-called FreqTransNet. By leveraging the capabilities of convolutional neural networks, a lightweight spatial transformer module, and frequency-domain transformations, FreqTransNet exhibits significant structural flexibility and effectively captures multi-scale features within images. Comprehensive experimental results demonstrate that FreqTransNet surpasses existing SOTA-based methodologies in terms of imperceptibility and robustness. Additionally, ablation studies validate the contributions of each component to the model's overall performance.

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